

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F

1. A crowbar used to pull a nail out of wood is an example of which simple machine? [1]

A screw

A lever

A gear

An inclined plane

A pulley

A wheel and axle

2. When building a robotic hand using cardboard, the fold or crease in the cardboard finger acts as which part of a lever? [1]

The pulley

The effort

The string

The wedge

The load

The fulcrum

3. A student pushes a box with a force of 10 Newtons for a distance of 5 meters. How much work is done? [1]

5 Joules

50 Joules

100 Joules

15 Joules

5 Joules

2 Joules

4. Why are sports cars designed with a wedge shape at the front? [1]

To use larger gears.

To look like a doorstep.

To increase the weight of the car.

To lift the car off the ground.

To cut through air particles (aerodynamics).

To carry more passengers.

5. Why do elevators (lifts) often use a counterweight? [1]

To increase the distance the elevator travels.

To make the elevator look modern.

To make the elevator fall faster.

To increase the friction on the cables.

To act as a wedge if the cable snaps.

Joules

To help balance the load so the motor requires less force.

Newtons

Watts

Grams

Meters

6. If you use a pulley system that reduces the force needed by half, what happens to the rope? [1]

You need to pull the rope a shorter distance.

You need to pull the rope a longer distance.

You need a thicker rope.

The rope ties itself in a knot.

You need a stronger rope.

The rope becomes heavier.

9. Mountain roads often use switchbacks (zig-zags) to drive up the mountain. What does this achieve? [1]

It makes the mountain shorter.

It creates a wedge shape.

It creates a gentler slope (shallow angle) requiring less force.

It increases the speed of the car.

It allows the car to use a smaller gear.

It acts as a lever for the car.

7. A drill bit is a type of screw. It changes turning force into which type of motion? [1]

Circular motion only

Backward motion

Random motion

Linear (forward) motion

Sideways motion

No motion

10. A box is pushed up a 5-meter long inclined plane with a force of 40 Newtons. How much work is done? [1]

8 Joules

45 Joules

400 Joules

35 Joules

200 Joules

2000 Joules

8. What is the standard unit of measurement for "Work"? [1]

Seconds